

Global Technology & Internet Adoption Dashboard

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Project Overview

This project analyzes global technology adoption, internet accessibility, connectivity, and ICT development using the World Development Indicators (WDI) dataset. The dashboard was built in Microsoft Power BI to provide interactive insights across countries, regions, income groups, and years.

Objectives

- Analyze global technology adoption trends from 1960–2025.
- Compare countries using key technology indicators.
- Explore technology differences by income group and region.
- Build an interactive dashboard for decision-making.

Dataset

The analysis uses the World Development Indicators (World Bank) dataset. Key indicators include Internet users, mobile cellular subscriptions, fixed broadband subscriptions, fixed telephone subscriptions, and ICT goods exports.

Data Preparation

Data was cleaned in Power Query by filtering relevant indicators, unpivoting yearly columns, removing null values, correcting data types, and creating relationships between country, series, and date tables.

Dashboard Components

- KPI Cards: Total Countries, Average Indicator Value, Latest Year, Total Indicators, Total Records.
- Line Chart: Global Technology Adoption Trend (1960–2025).
- Bar Chart: Top 10 Countries by Technology Indicator.
- Map: Technology Indicator Distribution Across Countries.
- Donut Chart: Countries by Income Group.
- Scatter Chart: Internet Usage vs ICT Goods Exports.
- Detail Table with country-level records.

Key Insights

1. Global technology adoption has accelerated significantly since the early 2000s.
2. High-income economies generally record the strongest technology indicators.
3. Technology adoption varies considerably across regions and income groups.
4. Interactive filtering enables focused analysis by year, country, region, income group, and indicator.

Recommendations

- Expand broadband infrastructure in underserved regions.
- Promote digital inclusion initiatives.
- Encourage ICT investment and innovation.
- Use interactive dashboards for continuous monitoring of digital development.

Tools Used

Microsoft Power BI, Power Query, DAX, World Development Indicators (WDI).

Conclusion

The dashboard provides a comprehensive and interactive view of global technology and internet adoption, demonstrating practical skills in data preparation, modeling, visualization, and business intelligence.